Remote Internship Program

Task 2 Report

Phishing Awareness Training — Interactive Web Module

Intern Report

April 28, 2026

1 Objective

The goal of this task was to design and develop an interactive phishing awareness training module targeting end users. The module educates users on how to recognize phishing emails and fake websites, understand social engineering tactics used by attackers, and apply best practices to protect themselves. The deliverable includes real-world case studies and an interactive quiz to reinforce learning.

2 Tools and Environment

Component	**Details**
Languages	HTML5, CSS3, JavaScript (Vanilla)
Editor	Visual Studio Code
Platform	Any modern web browser
Dependencies	None — fully self-contained single file
Hosting	GitHub (static HTML)

3 Background

Phishing is one of the most prevalent and damaging cyberattack vectors in use today. According to industry reports, over 90% of successful data breaches begin with a phishing email. Unlike technical exploits that target software vulnerabilities, phishing attacks exploit human psychology — making user education the most effective line of defense.

This training module was designed around the following threat categories:

- **Email Phishing** — mass deceptive emails impersonating trusted entities.
- **Spear Phishing** — targeted attacks using personal information.
- **Whaling** — spear phishing directed at executives or high-value targets.
- **Smishing** — phishing delivered via SMS text messages.
- **Vishing** — voice phishing conducted over phone calls.
- **Search Engine Phishing** — fake websites appearing in paid search results.

4 Module Structure

The training is organized into seven sequential modules, each covering a distinct aspect of phishing awareness:

Module	Title	Content Summary
01	What is Phishing?	Attack type taxonomy with descriptions
02	Recognizing Phishing Emails	Annotated interactive phishing email
03	Spotting Fake Websites	Live URL inspector tool
04	Social Engineering Tactics	Six psychological manipulation tactics
05	Real-World Case Studies	Four documented attacks and lessons
06	Best Practices & Defense	Eight actionable protection tips
07	Interactive Quiz	8-question scored assessment

5 Implementation

5.1 Module 02 — Interactive Phishing Email

A simulated phishing email was built directly into the module. Each suspicious element is tagged with a hoverable warning indicator. When the user hovers over a flag, a tooltip explains exactly why that element is a phishing signal. Elements covered include:

- Typosquatted sender domain (`paypa1.com` vs `paypal.com`)
- Generic greeting instead of personalized name
- Artificial urgency and deadline pressure
- Mismatched hyperlink — display text vs real destination URL
- Excessive punctuation and capitalization in subject line

5.2 Module 03 — URL Inspector Tool

An interactive URL analysis tool was implemented in JavaScript. Users can either type a URL or click pre-loaded example chips. The analyzer evaluates the input against a set of rules and returns one of three verdicts:

- **DANGER** — known phishing patterns detected (typosquatting, suspicious TLD, domain spoofing).
- **WARNING** — unknown domain, proceed with caution.
- **SAFE** — matches a known legitimate domain.

Detection patterns include typosquatting (e.g., `paypa1`), suspicious TLDs (`.ru`, `.tk`, `.xyz`), subdomain spoofing (e.g., `netflix.com.verify-account.xyz`), and unencrypted HTTP connections.

5.3 Module 04 — Social Engineering Tactics

Six core psychological manipulation tactics are presented, each with a real-world example quote and explanation:

Tactic	Mechanism
Urgency & Scarcity	Creates time pressure to bypass rational thinking
Fear & Intimidation	Threatens legal action, fines, or account loss
Too Good to Be True	Lures victims with prizes or unexpected rewards
Authority & Trust	Impersonates executives, banks, or IT departments
Pretexting	Fabricates a scenario to justify information requests
Reciprocity	Does something first, then requests something in return

5.4 Module 05 — Real-World Case Studies

Four documented phishing incidents are presented with their financial or operational impact and key lessons:

- **Google & Facebook (2016)** — $100M+ lost to a Business Email Compromise (BEC) attack impersonating a vendor through fake invoices over two years.
- **DNC Spear Phishing (2016)** — A fake Google security alert email led to credential theft affecting a US presidential election campaign.
- **Twitter Vishing Attack (2020)** — IT impersonation via phone calls compromised admin credentials, hijacking major public accounts for a Bitcoin scam.
- **MGM Resorts (2023)** — A 10-minute phone call using LinkedIn-sourced employee information caused $100M in losses and a week-long system outage.

5.5 Module 07 — Interactive Quiz

An 8-question scenario-based quiz was implemented in JavaScript with the following features:

- Each question presents a realistic phishing scenario with four answer choices.
- On selection, all options are immediately locked and color-coded (correct in green, incorrect in red).
- A detailed explanation is shown beneath each answer whether the user was right or wrong.

- A progress bar tracks advancement through the quiz.
- A final results screen displays the score, a performance grade, and personalized feedback.

Grading thresholds:

Score	Grade	Message
8/8	Expert Level	Perfect score
6–7	Well Prepared	Review missed questions
4–5	Needs Improvement	Review all modules
0–3	High Risk	Immediate full review required

6 Deployment

The module was deployed as a static HTML file to GitHub. No server or backend is required — the file runs entirely in the browser.

```
git init
git add phishing_training.html README.md
git commit -m "Add Task 2 - Phishing Awareness Training
   interactive module"
git branch -M main
git remote add origin https://github.com/Douaax/
   CodeAlpha_Phishing_Awareness-_Training.git
git push -u origin main
```

The repository is publicly accessible at:
`https://github.com/Douaax/CodeAlpha_Phishing_Awareness-_Training`

7 Results

The completed module successfully fulfills all five task requirements:

- **Presentation/Module** — A complete 7-module interactive web training module.

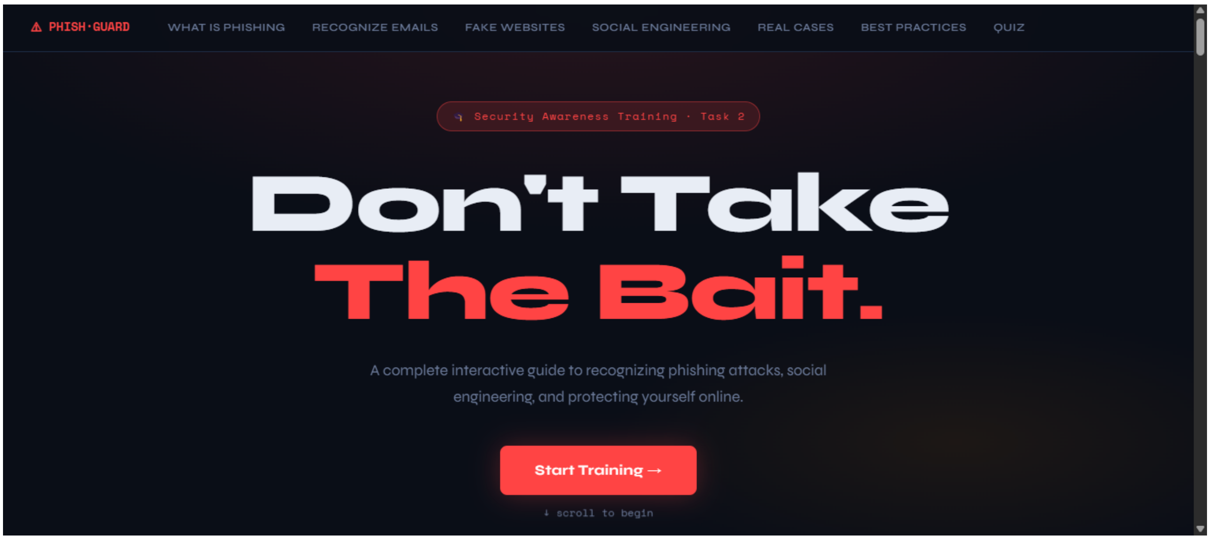

Figure 1: Landing Page

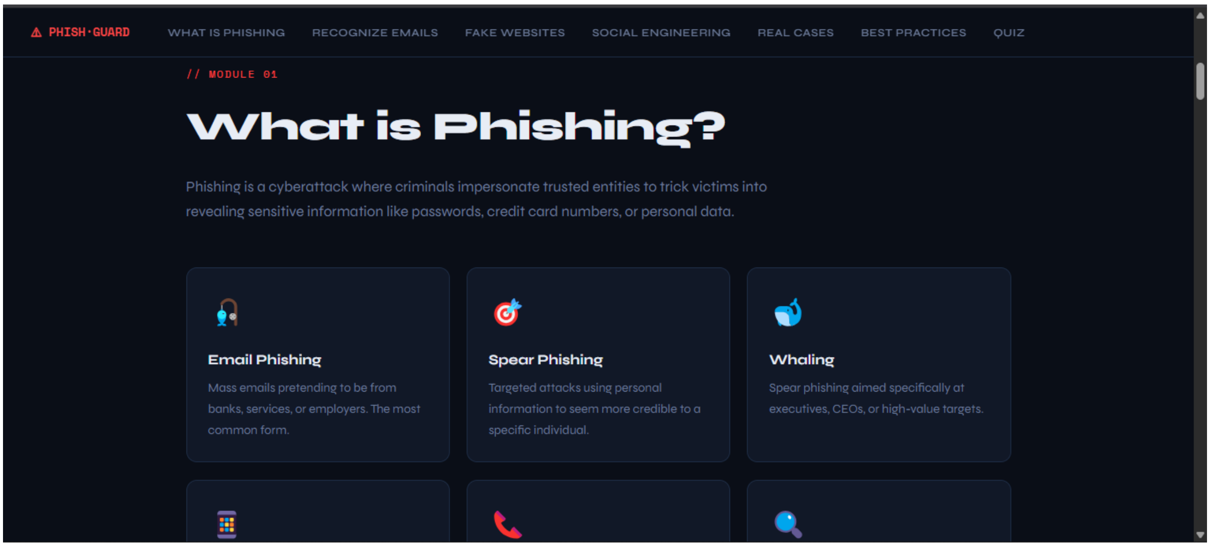

Figure 2: What is Phishing — Module 1

- **Recognize Phishing Emails** — Covered in Module 02 with a live annotated email.

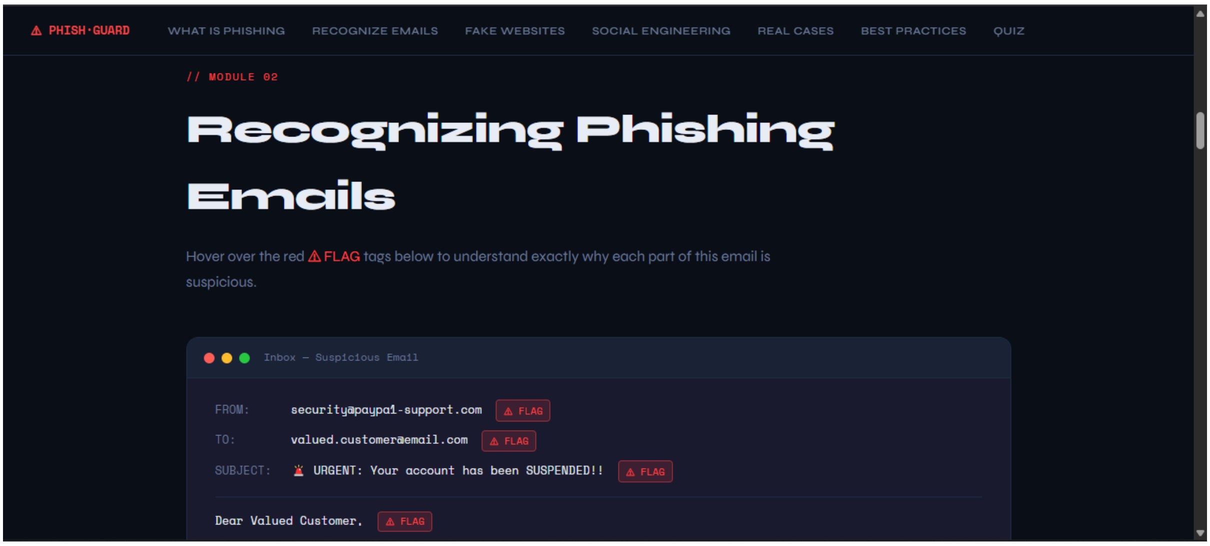

Figure 3: Recognize Phishing Emails — Module 2

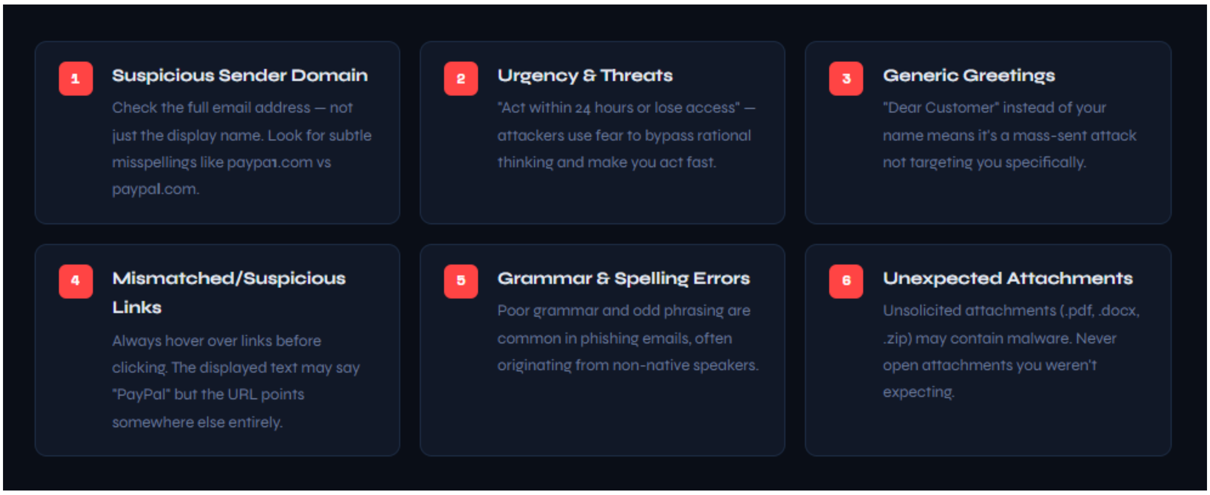

Figure 4: Recognize Phishing Emails 2 — Module 2

- **Fake Websites** — Covered in Module 03 with a functional URL inspector tool.

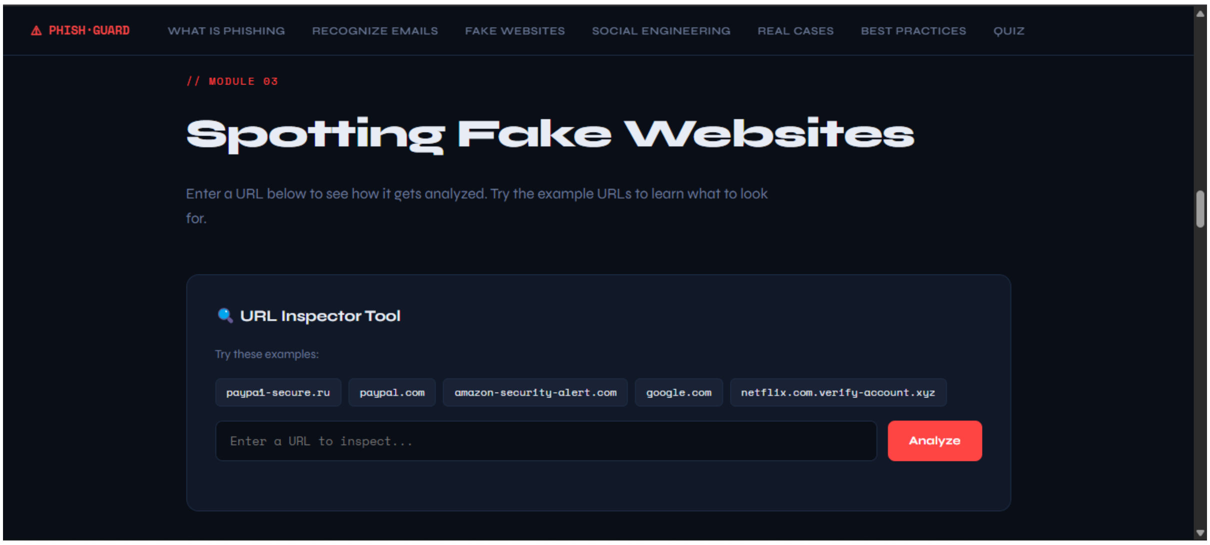

Figure 5: Fake Websites — Module 3

Figure 6: Spotting Fake Websites — Module 3

- **Social Engineering Tactics** — Covered in Module 04 with six documented tactics.

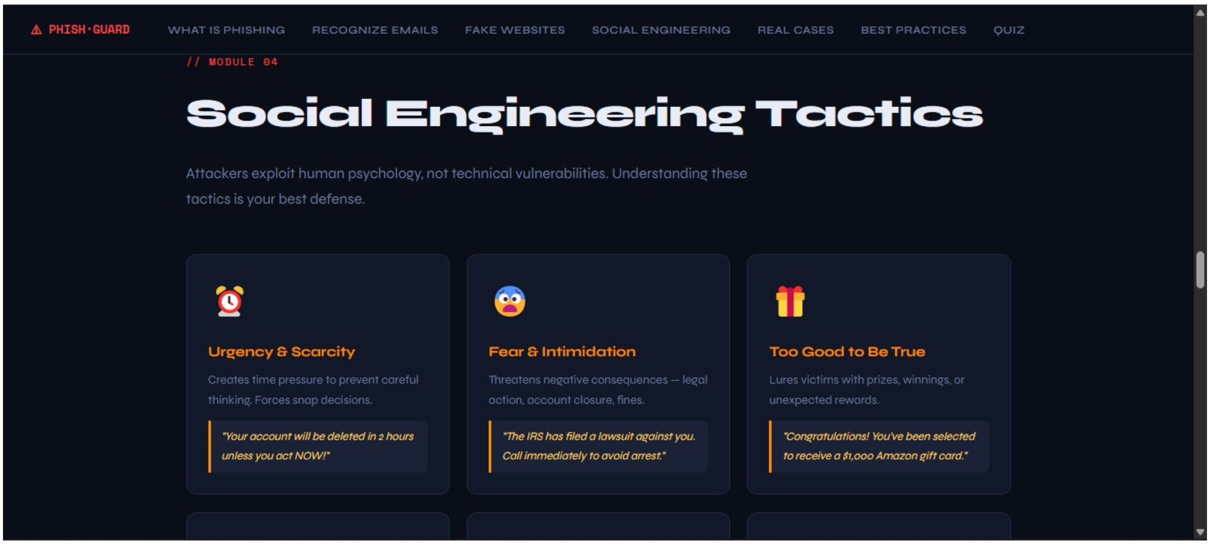

Figure 7: Social Engineering Tactics — Module 4

- **Best Practices** — Covered in Module 06 with eight actionable defense tips.

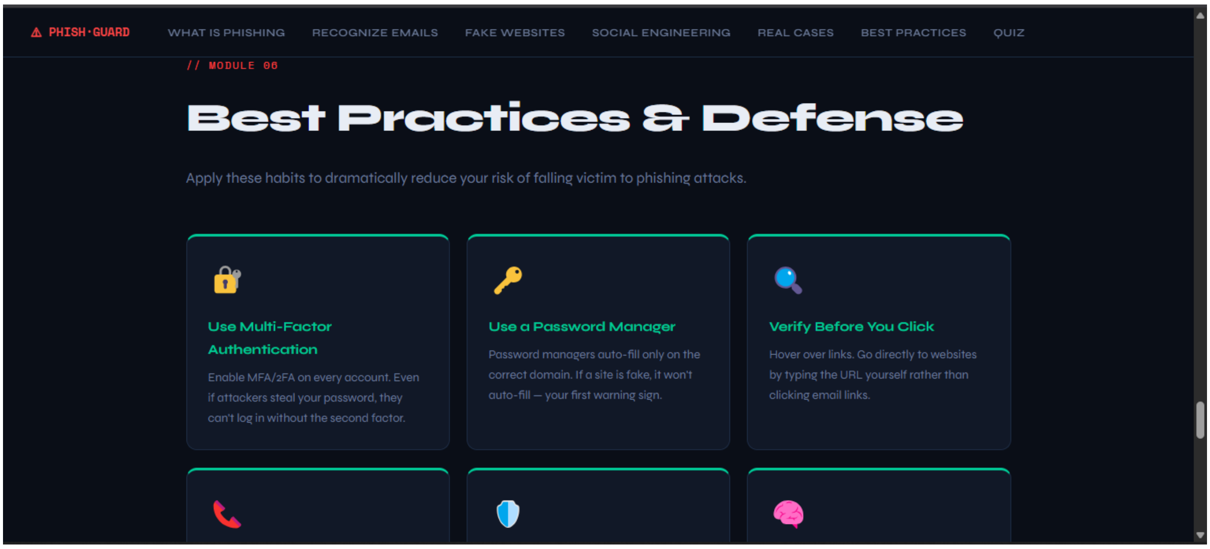

Figure 8: Best Practices — Module 6

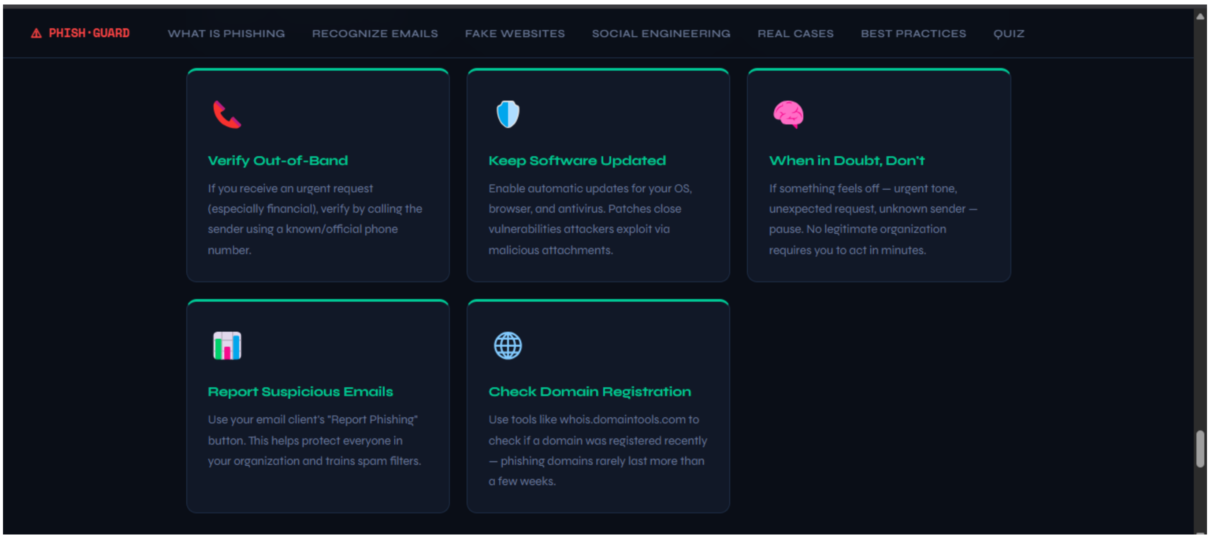

Figure 9: Best Practices and Defenses — Module 6

- **Real-World Examples & Interactive Quiz** — Covered in Modules 05 and 07.

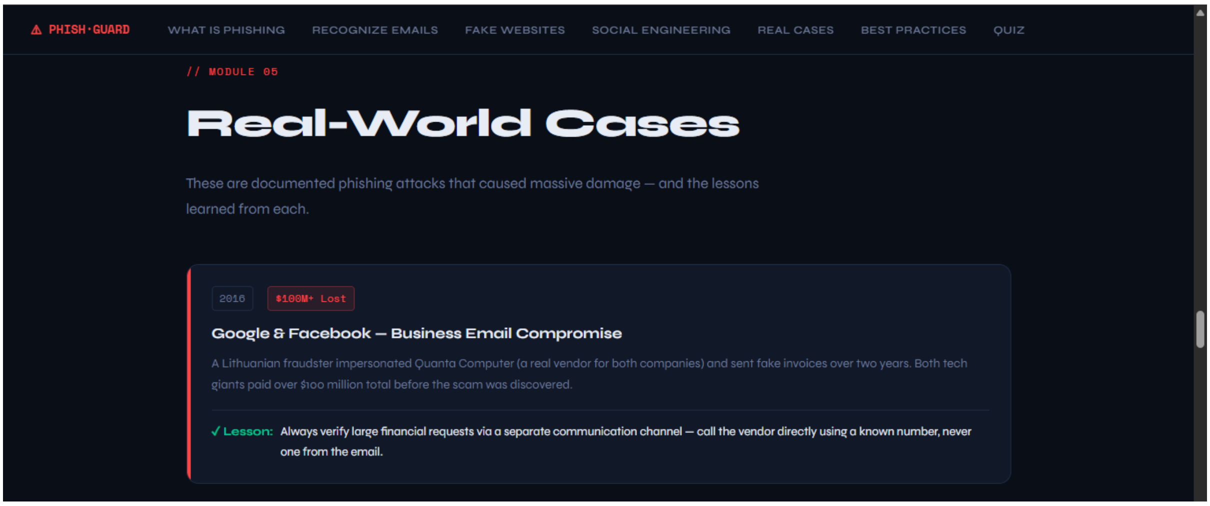

Figure 10: Real-World Examples — Module 5

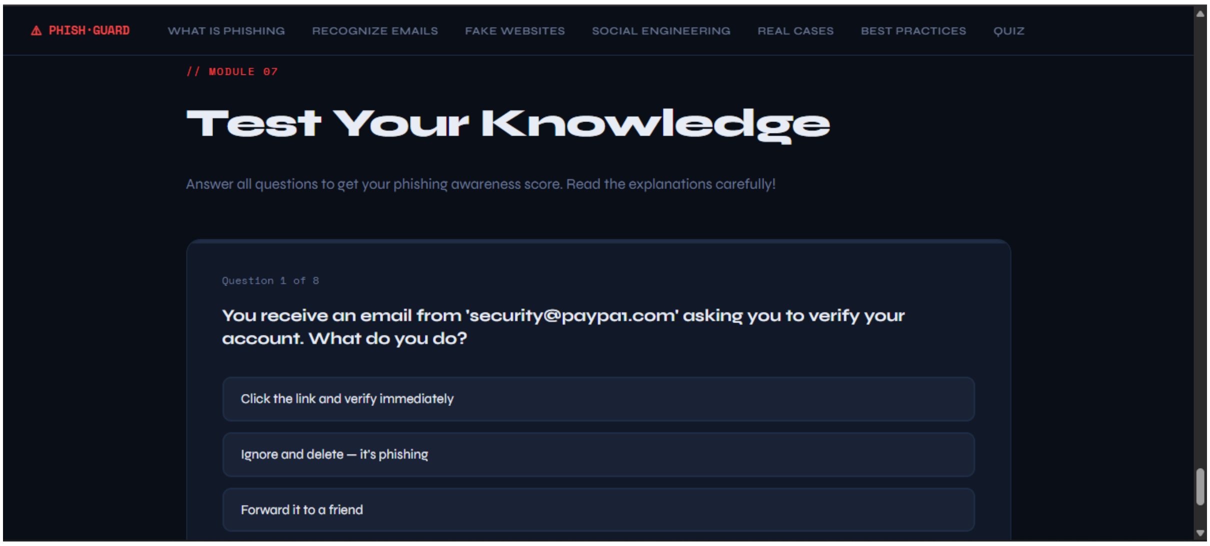

Figure 11: Test Your Knowledge — Module 7

8 Conclusion

This task produced a fully self-contained, interactive phishing awareness training module deployable in any browser without installation or dependencies. The module combines educational content with interactive tools and scenario-based assessment to maximize user engagement and knowledge retention. The quiz component in particular ensures that users actively process the material rather than passively reading it, reinforcing the critical security behaviors needed to resist real-world phishing attacks.